

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Information and communications technology

Software Requirement Specification

Version 1.2

AIMS System

Subject: ITSS Software Development

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1 Introduction

1.1 Objective

The main objective of this Software Requirements Specification (SRS) document is to provide a comprehensive architectural overview and detailed functional/non-functional requirements for the AIMS (An Internet Media Store) software. This document serves as the primary reference for the development team, software architects, software testers, and project managers to design, implement, and verify the system objectives

1.2 Scope

□ **Name:** AIMS (An Internet Media Store).

□ **Brief Description:** AIMS is a 24/7 e-commerce application (website or desktop app) designed to facilitate the buying and selling of physical media products, initially including books, newspapers, CDs, and DVDs.. It allows customers to browse, cart, and purchase items without logging in, while providing secure administrative portals for Product Managers and Administrators to manage inventory, orders, and staff roles.

□ **Goals & Benefits:** The system aims to provide a highly available and fast shopping experience. It is engineered to serve up to 1,000 customers simultaneously without significant performance degradation, operate continuously for 300 hours, and ensure a maximum response time of 2 to 5 seconds.

1.3 Glossary

No	Term	Explanation	Example	Note
1	AIMS	An Internet Media Store.	N/A	The core software product.
2	VAT	Value-Added Tax.	10% VAT	Excluded from the original value and current price.
3	Physical Media	Tangible media products supported by the store.	Books, CDs, DVDs, Newspapers	Other types may be supported in the future.

4	PayPal Sandbox	A testing environment for credit card payment integration.	Mock credit card transaction	Used for demonstration purposes.
5	VietQR	An API for generating and processing QR code payments.	VietQR Sandbox	The default payment method.

1.4 References

AIMS: An Internet Media Store - Problem Statement v3.1 (Lecturer: NGUYEN Thi Thu Trang)

2 Overall Description

AIMS is an e-commerce platform designed with simplicity and performance in mind. It separates the customer-facing shopping experience from the internal management workflows. To streamline purchasing, the customer interface requires no authentication. In contrast, the internal system requires role-based access control for administrative and catalog management tasks

2.1 Survey

List of Actors and Descriptions:

- **Customer** A user who browses the catalog, manages their shopping cart, and places orders.
- **Product Manager** A staff member responsible for maintaining the product catalog (CRUD operations, stock adjustments) and reviewing (approving/rejecting) pending orders.
- **Administrator:** A staff member who manages internal user accounts, assigns roles, and handles security actions like blocking users or resetting passwords.
- **User:** A generalized internal staff member (parent of Administrator and Product Manager) who must log in to access the system.
- **Payment System:** External systems (VietQR System and PayPal Sandbox System) that process transaction requests and return payment statuses.

2.2 Overall requirements

2.2.1. Customer

- **Browse Products:** The customer browses the catalog, searches, and filters available media products within the system.
- **View Product Detailed Information:** The customer views the specific details of a particular product.
- **Check Cart:** The customer reviews their shopping cart (this action inherently includes Display Cart Info). During this process, the customer can optionally execute extending Use Cases to Change quantity of products or Remove from cart.
- **Place order:** The customer proceeds to checkout. This process mandates the customer to provide a delivery address via Set up Delivery Info and complete the payment via Pay Order.
- **View Order Info & Cancel Order:** The customer tracks the order status post-purchase and can request to cancel the order if necessary.

2.2.2. Product Manager

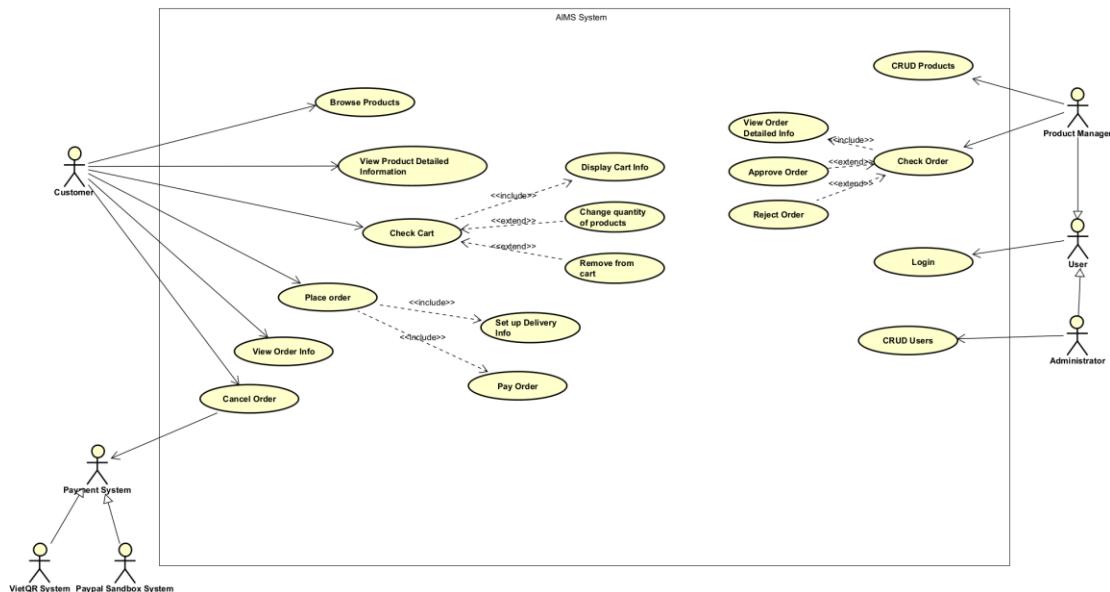
- **Check Order:** The manager accesses the queue of pending orders (automatically including View Order Detailed Info to display the specifics). Based on this review, the manager can optionally extend the flow to Approve Order (to process fulfillment) or Reject Order (to cancel it).
- **CRUD Products:** The manager handles the lifecycle of physical media products in the system, which includes creating new entries, updating information, adjusting stock levels, and deleting products.

2.2.3. Administrator

- **CRUD Users:** The administrator is responsible for managing internal staff accounts, which includes granting permissions, blocking/unblocking accounts, and resetting passwords.

2.2.4. User (Abstract Staff Member)

- **Login:** This is a mandatory Use Case for any staff member (including Product Managers and Administrators) to authenticate and access the internal management portals of the AIMS system.



2.3 Business process

2.3.1. Purchasing Process

- **Step 1:** The Customer uses Browse Products to find desired items and selects View Product Detailed Information to examine the specifics.
- **Step 2:** The Customer uses Check Cart to review the selected items. If modifications are needed, the Customer triggers the extending Use Cases: Change quantity of products or Remove from cart.
- **Step 3:** To finalize the purchase, the Customer selects Place order.
- **Step 4:** The system sequentially requires the Customer to execute Set up Delivery Info to calculate shipping fees.
- **Step 5:** The Customer completes the process by executing Pay Order via the default payment gateway (VietQR System) or an alternative method (PayPal Sandbox System).

2.3.2. Order Fulfillment Process

- **Step 1:** The Product Manager performs Login to enter the management portal.
- **Step 2:** The Product Manager selects Check Order to retrieve the list of newly placed orders (the system displays details via View Order Detailed Info).
- **Step 3:** The Product Manager evaluates the inventory conditions and decides to execute either Approve Order (to proceed with packaging and shipping) or Reject Order (to cancel the order, which subsequently triggers a refund via the Payment System).

2.3.3. Data Management Process

- **Product Catalog Management:** Whenever new inventory arrives or pricing adjustments are required, the Product Manager will Login and use CRUD Products to update the catalog database.
- **Personnel Management:** When a new employee is onboarded or an existing one departs, the Administrator will Login and use CRUD Users to provision a new account, assign appropriate roles, or revoke access rights.

3 Detailed Requirements

<This section contains all the software requirements at a level of detail sufficient to enable designers to design a system to satisfy those requirements, and testers to test that the system satisfies those requirements. Throughout this section, every stated requirement should be externally perceivable by users, operators, or other external systems. These requirements should include at a minimum a description of every input (stimulus) into the system, every output (response) from the system and all functions performed by the system in response to an input or in support of an output. The following principles apply:

- (1) Specific requirements should be stated with all the characteristics of a good SRS*
 - *correct*
 - *unambiguous*
 - *complete*
 - *consistent*
 - *ranked for importance and/or stability*
 - *verifiable*
 - *modifiable*
 - *traceable*
- (2) Specific requirements should be cross-referenced to earlier documents that relate*
- (3) All requirements should be uniquely identifiable (usually via numbering like 3.1.2.3)*
- (4) Careful attention should be given to organizing the requirements to maximize readability (Several alternative organizations are given at end of document)*

Before examining specific ways of organizing the requirements it is helpful to understand the various items that comprise requirements as described in the following subclasses. This section reiterates section 2, but is for developers not the customer. The customer buys in with section 2, the designers use section 3 to design and build the actual application.

Remember this is not design. Do not require specific software packages, etc unless the customer specifically requires them. Avoid over-constraining your design. Use proper terminology:

The system shall... A required, must have feature

The system should... A desired feature, but may be deferred til later

The system may... An optional, nice-to-have feature that may never make it to implementation.

Each requirement should be uniquely identified for traceability. Usually, they are numbered 3.1, 3.1.1, 3.1.2.1 etc. Each requirement should also be testable. Avoid imprecise statements like, “The system shall be easy to use” Well no kidding, what does that mean? Avoid “motherhood and apple pie” type statements, “The system shall be developed using good software engineering practice”

Avoid examples, this is a specification, a designer should be able to read this spec and build the system without bothering the customer again. Don’t say things like, “The system shall accept configuration information such as name and address”. The designer doesn’t know if that is the only two data elements or if there are 200. List every piece of information that is required so the designers can build the right UI and data tables>

3.1 Use case 1

Use Case “View Product Detail”

1. Use case code

UC005

2. Brief Description

This use case describes the interaction between the Customer / Product Manager and the AIMS system when the actor wants to view the detailed information of a product that has already been displayed in the product list.

3. Actors

3.1 Name of Actor 1

Customer

3.2 Name of Actor 2

Product Manager

4. Preconditions

The AIMS system is operating normally, and the product already exists in the system. The actor is currently on the product list screen, search results screen, or product management screen. In the case of the Product Manager, the actor has logged in and has permission to access the product management function. The selected product may be either available for sale or currently under management in the system.

5. Basic Flow of Events

1. The actor accesses the product list.
2. The system displays the product list.
3. The actor selects a product to view its details (**see Table A**).

4. The system receives the request to view the product details.
5. The system retrieves the detailed information of the selected product from the database.
6. The system displays the general information of the product, including barcode, title, category, original value, current price, general description, height, width, length, weight, quantity in stock and availability status (**see Table B**).
7. The system checks the product type.
8. The system displays additional type-specific information (**see Table B**)
9. If the actor is a Customer, the system also displays the **Add to Cart** option and the quantity selection (**see Table B**).
10. The use case ends when the actor finishes viewing the product details.

6. Alternative flows

Table 1 - Alternative flows of events for UC ViewProductDetail

No	Location	Condition	Action	Resume location
1.	At Step 3	If the actor chooses to return the product list	▪ The system returns to the product list screen	Use case ends
2.	At Step 5	If the product does not exist or has been removed from the system	▪ The system notifies that the product does not exist or its details cannot be viewed	Use case ends
3.	At Step 5	If the detailed product data is missing or contains errors	▪ The system displays the available information and notifies that some detailed information is currently unavailable	Resumes at Step 10
4.	At Step 8	If the product type is unknown or not supported	▪ The system displays only the general information of the product	Resumes at Step 10
5.	At Step 9	If the actor is a Product Manager	▪ The system does not display the Add to Cart function and only allows the actor to view the information	Resumes at Step 10
6.	At Step 9	If the Customer selects Add to Cart	▪ The system proceeds to the Add Product to Cart use case	Use case ends

7. Input data

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Product ID	The identifier of the product selected for viewing details	Yes	Must exist in the system	A12345

8. Output data

No	Data fields	Description	Display format	Example
1.	Title	Product title	Text	One piece
2.	Barcode	Product barcode	Numeric string	012345678
3.	Category	Product category	Text	Dramatic
4.	Original price	The original value of the product (without 10% VAT)	Currency(VNĐ)	123.456 VNĐ
5.	Current price	The current selling price of the product (displayed without 10% VAT)	Currency(VNĐ)	234.567 VNĐ
6.	Description	General description of the product	Paragraph text	New book,...
7.	Dimensions	Product dimensions	Text	20x15x10 cm
8.	Weight	Product weight	Text	1 kg
9.	Stock information	Quantity in stock, availability status	Interger	150
10.	Author(s)	Author(s) of the book	Text	Nguyễn Ái Quốc
11.	Cover type	Type of book cover	Text	Paperback
12.	Publisher	The publisher of the product.	Text	NXB Kim Đồng
13.	Publication date	Date of publication	Date	15/03/2026
14.	Number of pages	Total number of pages	Number	123
15.	Language	Language of the product	Text	Vietnamese
16.	Genre	The genre of product	Text	Comedy/ Action
17.	Editor-in-chief	Editor-in-chief of newspaper	Text	Lê Anh Đạt
18.	Issue number	Issue number of product	Number	123
19.	Publication frequency	Publication frequency of product	Text	Weekly

20.	ISSN	ISSN code	Text	1234-5678
21.	Sections	Newspaper sections	Text	Business
22.	Artists	The artists of product	Text	Jack 97
23.	Record label	Record label of CD	Text	MTP entertainment
24.	Track list	List of tracks	List	Track 1 – 2:34
25.	Disc type	Type of disc	Text	Blu – ray
26.	Director	Director of product	Text	Son Tung MTP
27.	Runtime	The runtime of product	Text	60 minustes
28.	Studio	The production studio of the DVD	Text	CDA
29.	Subtitles	Subtitles of product	Text	Vietnamese
30.	Type	Type of product	Text	DVD

9. Postconditions

- The actor has viewed the detailed information of the selected product.
- No system data is changed if the actor only views the product details.
- The system preserves the current state of the product list, search results so that the actor can return and continue interacting.

3.2 Use case 2

Use Case “Place Order”

1. Use case code

UC002

2. Brief Description

This use case describes the interaction between Customer and Aims system when Customer places an order for products in the cart and proceeds to payment.

3. Actors

- a) Customer

4. Preconditions

- a) The Aims system is running normally
- b) Products in the cart exist in the system
- c) There is at least one item in the cart.

5. Basic Flow of Events

1. Customer requests to place an order from the cart
2. The system checks whether the available stock is sufficient to fulfill the customer’s demands
3. If the stock is sufficient, the system displays the delivery information form together with the cart information
4. Customer enters and submits delivery information

5. The system validates the delivery information and asks the customer to update if in case any required field is blank or invalid
6. Each time the customer enters or updates the delivery address, the system calculates or recalculates the shipping fee
7. The system displays and temporarily saves the invoice information, including the list of products in the cart, quantity, product prices, total product price excluding VAT, total product price including VAT, delivery fee, and total amount need to be paid
8. Customer requests to pay for the order
9. The system continues with the use case Pay Order
10. After successful payment, the system displays the general order information, including customer name, phone number, shipping address, province, and total amount, together with transaction information including transaction ID, transaction content, and transaction time
11. The system clears the cart
12. The system sets the order status to Pending Processing
13. The system sends the invoice and payment transaction information to the customer's email
14. The system records the payment transaction and successful order information

6. Alternative flows

Table N-Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
1.	At Step 2	One or more products in the cart do not have sufficient stock to fulfill the customer's requested quantity	The AIMS software notifies the customer that some products in the cart are out of stock or insufficient. Customer is required to update the cart	Use case ends and returns to View Cart
2.	At Step 5	The delivery information entered by the customer is incomplete	The system displays an error message indicating the invalid or missing information. The customer is requested to re-enter valid delivery information	Step 3: Display delivery information form)
3.	At any step	The customer want to go back to a previous screen or exits the ordering process	The system saves the current cart information The ordering process is canceled or paused The ordering process is canceled or paused	Use case ends
4.	At step 9	Payment fails or the customer cancels the payment while it is in process	The system displays a message that Payment is unsuccessful The customer may retry the payment or modify the order	Step 7: displays invoice information

7. Input data

Table A-Input data of place order

No	Data fields	Description	Mandatory	Valid condition	Example
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1.	String	Name of the person who will receive the order	Yes	Not empty	Bùi Doãn Khang
2.	Numeric	Phone number of the receiver used for delivery contact	Yes	- Not empty - Contain 10 digits and start with '0'	0123456789 0123 456 789
3.	String	Province or city where the order will be delivered	Yes	Must be selected from the list of supported provinces/cities in Vietnam	Hà Nội Hồ Chí Minh city
4.	String	Detailed address where the order will be delivered	Yes	Not Empty	1 trần đại nghĩa street, hà nội
5.	String	Some notes for the delivery process.	No	Not exceed 255 characters	Please call before delivery or send message

8. Output data

Table B-Output data of Place Order

No	Data fields	Description	Display format	Example
1.	String	Title of the media product displayed in the invoice	Text	Books: master machine learning
2.	Integer	Price of a product in the order (excluding VAT)	Positive number with thousands separator	120,000
3.	Integer	Quantity of the product in the order	Positive integer	2
4.	Integer	Total price for each product (Product Price × Quantity)	Positive number with thousands separator	1,000,000
5.	Integer	Total price of all products in the order before VAT is applied	Positive number with thousands separator	5,000,000
6.	Integer	Total product price including VAT	Positive number with thousands separator	5,200,000
7.	Integer	Shipping fee calculated by the system based on delivery address and product weight	Positive number with thousands separator	40,000
8.	Integer	Total payment amount including product price (include VAT) and delivery fee	Positive number with thousands separator	5,240,000
9.	String	Name of the person receiving the order	Text	Bùi Doãn Khang
10.	Integer	Phone number of the order receiver	Numeric Text	0123456789
11.	String	Delivery address of the order	Text	1 trần đại nghĩa street, hà nội

12.	String	Province or city of the delivery address	Text	Hà Nội
13.	String	ID of the payment Transaction	Numeric Text or/and Text	ABC123
14.	String	Description or content of the payment transaction	Text	Payment for ABC123
15.	Datetime`	Date and time when the payment transaction occurred	dd/MM/yyyy HH:mm:ss	1/1/2026 12:30:00

9. Postconditions

Success Postcondition:

- The place order state is successful
- A new order is created in the system
- The order status is set to Pending Processing
- The system records the order information and payment transaction information
- The customer's cart is cleared
- The system sends the invoice and payment transaction information to the customer's email

Failure Postcondition

- No order is created in the system
- No payment transaction is recorded
- The products remain in the customer's cart so that the customer can modify the order or attempt payment again

3.3 Use case 3

Use Case “Pay Order (VietQR)” 1. Use case code: UC003 2. Brief Description: This use case involves the integration of payment using the VietQR method, allowing customers to pay for physical media products (books, newspapers, CDs, DVDs) via a QR code. 3. Actors: • Customer: The user purchasing physical media. • VietQR Service: The external system providing the API for QR generation and transaction sync. 4. Preconditions • The system has calculated the current price of products. • The system has created connection to VietQR. • The customer has placed an order for the products in their cart with delivery information.	
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5. Basic Flow of Events

1. The customer selects QR code as the payment method for their order.
2. The system calls the VietQR Payment API and send information(See Table A).
3. VietQR Payment generate a dynamic QR code based on the order details and send to the system(See Table B).
4. The system displays the generated QR code to the customer.
5. The customer completes the payment using the QR code.
6. VietQR system handle the transaction and send transaction information to system.
7. The system returns the payment transaction result and information to the calling use case ("Place Order").

6. Alternative flows:

No	Location	Condition	Action	Resume location
1.	At Step 2	If the system cannot connect to the VietQR system	▪ The system displays error message.	Use case ends
2.	At Step 4	If QR code is invalid or expired	▪ The system displays the message "Payment failed or overdue". ▪ The system takes the customer back to the order page.	Use case ends
3.	At Any Step	If the customer wants to cancel the payment transaction(e.g. customer wants to	▪ The system displays the message "Cancelled Transaction".	Use case ends

		change payment method)	The system takes the customer back to the order page	
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7. Input data

Table B

No	Data fields	Description	Mandatory	Valid condition	Example
1	QR Link	The URL contains a QR code image automatically generated by the VietQR system.	Yes	Must be a valid HTTPS URL	https://img.vietqr.io/image/mb-1234567890-compact.png?amount=500000&addInfo=PAY98765

8. Output data

Table A

No	Data fields	Description	Display format	Example
1.	BankCode	Recipient's bank	Required, String	MB
2.	BankAccount	Recipient's bank account	Required, String (Numeric characters only)	99991112345
3.	UserBankName	Recipient name	Required, String	NGO DUC HUY
4.	TotalAmount	The actual amount of the order	Required, Integer, greater than 0	150.000
5.	Content	The transfer details will be visible to the customer.	Maximum 19 characters, written continuously without spaces or accents, and no special characters.	ORD12345
6.	OrderId	Transaction code for system to manage.	Required, String, Unique	ORD987654321

9. Postconditions

- The payment transaction is authorized and confirmed by the VietQR system. Transaction details (including Transaction ID, Transaction Datetime, and success status) are returned to the Use Case (Place Order) to continue the order recording flow.
- The transaction was cancelled, the QR code expired, or the payment failed. A failure/cancellation status is returned to the Use Case Place Order for the system to process the next step.

3.4 Use case 4

Use Case “Pay Order By Credit Card”

1. Use case code

UC004

2. Brief Description

This use case describes the interaction between the Customer, Paypal Sandbox System and AIMS System when the Customer wishes to pay for an order using a credit card.

3. Actors

3.1 Customer

3.2 Paypal Sandbox System

4. Preconditions

Customer in the Check Out page and choose the Credit Card Payment method

5. Basic Flow of Events

1. The Customer selects the "Credit Card" payment option and clicks "Proceed to Payment".
2. The software displays the credit card payment form.
3. The Customer enters their credit card details (see Table A) and clicks "Submit Payment".
4. The software validates the input data locally and sends a payment authorization request to the PayPal Sandbox API.
5. The PayPal Sandbox processes the simulated transaction and returns a successful response.
6. The software receives successful response.

6. Alternative flows

Table N-Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
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7.	At Step 4	If the input data is incomplete or violates validation rules (e.g., invalid card length, expired date).	- Software highlights the invalid fields. - Software displays a specific error message (e.g., "Invalid card number").	Resumes at Step 3
8.	At Step 4	If the software fails to connect to the PayPal Sandbox API (timeout or server error).	- Software logs the system error. - Software displays a "Gateway connection error. Please try again later." message.	Use case ends
9.	At Step 5	If the PayPal Sandbox API declines the test card (e.g., insufficient funds test card).	- Software logs the failed transaction attempt. - Software displays a "Payment Declined" error message.	Resume at Step 2

7. Input data

Table A - Input data of Credit Card

No	Data fields	Description	Mandatory	Valid condition	Example
2.	Card Number	The 16-digit number on the front of the card.	Yes	Exactly 16 numeric digits. Must pass Luhn algorithm check.	4111 1111 1111 1111
3.	Expiration Date	The month and year the card expires.	Yes	Format MM/YY. Must be a future date.	03/30
4.	CVV	Card Verification Value (security code).	Yes	Exactly 3 or 4 numeric digits.	123

8. Output data

9. Postconditions

The software successfully record Payment Transaction and Order Information

3.5 Use case 5

<Use case specification of use case 1 following the bellow template. You may need to design activity diagrams if the flow of events is complex>

Use case “CUD Product”

I. Use Case: Create Product

1. Use case code:

UC005-Create

2. Brief Description:

This use case describes the interaction between the Product Manager and the AIMS system when the actor wants to add a new product to the system. The Product Manager can add an unlimited number of products per day, but only one product at a time. The system validates the input data and ensures that all business rules are satisfied before completing the operation.

3. Actors:

Product Manager

4. Preconditions:

- The Product Manager needs to log in to access the features corresponding to their roles.
- The Product Manager is currently on the product management screen.

5. Basic Flow of Events:

1. The Product Manager accesses the product management screen.
2. The Product Manager chooses to add a new product.
3. The system displays a product creation form.
4. The Product Manager enters the essential core information and physical attributes of the product.
5. The Product Manager selects the product category (books, newspapers, CDs, or DVDs) and provides further details depending on the type of product.
6. The Product Manager submits the request.
7. The system validates the input data to ensure it complies with the rules.
8. The system stores the new product information in the database.
9. The software stores a history of products addition.
10. The system displays a success notification.
11. The use case ends.

6. Alternative flows:

- **Invalid Input:** The software will notify the product manager if any operation is invalid. For example, when the input does not comply with the rules or the date format is incorrect. Resume at Step 4 or 5.
- **Price Constraint Violation:** The price of the product must always be between 30% and 150% of the product's original value to avoid price inflation or undercutting. If violated, the system displays an error message. Resume at Step 4.

7. Input data:

No	Data fields	Description	Mandatory	Valid condition	Example
1	Barcode	Core detail: Unique identifier	Yes	Must not duplicate existing barcode	893523451234
2	Title	Core detail: Product title	Yes	Non-empty text	One Piece Vol.1
3	Category	Core detail: books, newspapers, CDs, DVDs	Yes	Must belong to supported types	Book
4	Original value	Core detail: Original value without a 10% value-added tax (VAT)	Yes	Must be positive number	100000
5	Current price	Core detail: Current price without a 10% value-added tax (VAT)	Yes	Between 30% and 150% of original value	120000

6	Description	General description (e.g., new or used condition, primary color, or return condition)	Yes	Text	Brand new, sealed
7	Dimensions	Physical attributes: Height, width, length	Yes	Valid measurement	20x15x5 cm
8	Weight	Physical attribute	Yes	Positive number	0.5 kg
9	Book details	Author(s), cover type (paperback or hardcover), publisher, publication date. Additional: number of pages, language, genre.	Yes (if Book)	Follow product rules	Author: Eiichiro Oda, Cover: Paperback, Publisher: Shueisha
10	Newspaper details	Editor-in-chief, publisher, publication date. Additional: issue number, publication frequency, ISSN, language, sections.	Yes(if Newspaper)	Follow product rules	Editor: John Doe, Frequency: Daily, Section: Sports
11	CD details	Artists, record labels, tracks list (title and length), genre. Additional: release date.	Yes (if CD)	Follow product rules	Artist: The Beatles, Genre: Rock
12	DVD details	Disc type (Blu-ray, HD-DVD), director, runtime, studio, language, subtitles. Additional: release date, genre.	Yes (if DVD)	Follow product rules	Type: Blu-ray, Director: James Cameron, Runtime: 192 mins

8. Output data:

No	Data fields	Description	Display format	Example
1	Operation result	Notification of successful product addition.	Text message	Product added successfully
2	Error / Validation message	Notification if the input does not comply with the rules or date format is incorrect.	Text message	Invalid date format / Current price must be at least 30% of original value

9. Postconditions:

- The product information is successfully created in the system.
- The software stores a history of products addition.
- The Product Manager can continue adding other products (unlimited per day).

II. Use Case: Update Product

1. Use case code

UC005-Update

2. Brief Description:

This use case describes the interaction between the Product Manager and the AIMS system when the actor wants to edit the information of an existing product. The system validates the input data and ensures that all business rules are satisfied before completing the operation.

3. Actors:

Product Manager

4. Preconditions

- The Product Manager has logged into the system to access the features corresponding to their role.
- The selected product must already exist in the system.

5. Basic Flow of Events

1. The Product Manager accesses the product management screen.
2. The system displays the list of products.
3. The Product Manager chooses to edit the information of an existing product.
4. The Product Manager selects only one product at a time to edit.
5. The system retrieves and displays the current product information.
6. The Product Manager modifies the required information (including stock adjustment and explicit reason if any) and submits the changes.
7. The system validates the input data to ensure it complies with the rules.
8. The system saves the updated product information in the database.
9. The software stores a history of product editing.
10. The system displays a success notification.
11. The use case ends.

6. Alternative flows

- **Invalid Input:** The software will notify the product manager if any operation is invalid. For example, when the input does not comply with the rules or the date format is incorrect. Resume at Step 6.
- **Price Constraint Violation:** The price of the product must always be between 30% and 150% of the product's original value to avoid price inflation or undercutting. If violated, the system displays an error message. Resume at Step 6.

7. Input data

No	Data fields	Description	Mandatory	Valid condition	Example
1	Title	Product title	Yes	Non-empty text	Dragon Ball
2	Category	Product category (books, newspapers, CDs, DVDs)	Yes	Must belong to supported types	newspaper
3	Original value	Original value without a 10% value-added tax (VAT)	Yes	Must be positive number	200000
4	Current price	Current selling price without a 10% value-added tax (VAT)	Yes	Between 30% and 150% of original value	210000
5	Description	General description (e.g., new or used condition, primary color, or return condition)	Yes	Text	slight wear
6	Dimensions	Height, width, length	Yes	Valid measurement	20x15x15 cm
7	Weight	Physical attribute	Yes	Positive number	0.6 kg
8	Quantity in stock	Number of items in stock	Yes	Integer ≥ 0	120

9	Reason for stock change	Explicitly recorded reason when stock needs to be adjusted manually	Yes (if stock is changed manually)	Non-empty text	Removed 2 damaged items
10	Product type specific info	Details depending on the type of product	Yes	Must follow product rules	Artist: The Beatles, Genre: Rock

8. Output data:

No	Data fields	Description	Display format	Example
1	Operation result	Notification of successful product update.	Text message	Product successfully updated
2	Error / Validation message	Notification explaining why an operation is invalid.	Text message	The price must be between 30% and 150% of the original value
3	Updated product information	Updated product data displayed after operation.	Structured text	Quantity in stock: 150

9. Postconditions:

- The product information is successfully updated in the system.
- The software stores a history of product editing.
- The product list is updated to reflect the latest changes.

III. Use Case: Delete Product

1. Use case code:

UC005-Delete

2. Brief Description:

This use case describes the interaction between the Product Manager and the AIMS system when the actor wants to delete products. The system checks simultaneous deletion limits, daily deletion limits, and stock quantities before performing the deletion or changing the product status to deactivated.

3. Actors:

Product Manager

4. Preconditions:

- The Product Manager needs to log in to access the features corresponding to their roles.
- The Product Manager is currently on the product management screen.
- The selected product(s) must already exist in the system.

5. Basic Flow of Events:

1. The Product Manager accesses the product management screen.
2. The Product Manager selects up to ten products to delete simultaneously.
3. The Product Manager submits the request to delete the selected products.
4. The system validates the daily deletion limit to ensure the manager does not delete more than twenty products per day.
5. For each selected product, the system checks if its stock is equal to zero.
6. Confirming the stock is zero, the system deletes the product(s).
7. The software stores a history of products deletion.
8. The system displays a success notification.
9. The use case ends.

6. Alternative flows:

- **Exceed Simultaneous Limit:** If the manager selects more than ten products at a time, the software will notify the product manager that the operation is invalid. Resume at Step 2.
- **Exceed Daily Limit:** If the deletion request causes the manager to exceed deleting twenty products per day, the system rejects the operation for security reasons and notifies the manager. Use case ends.
- **Stock Greater Than Zero:** If a product's stock is not equal to zero, it cannot be deleted. Instead, its status will be changed to "deactivated," meaning it will no longer be available for sale. Resume at Step 7.

7. Input data

No	Data fields	Description	Mandatory	Valid condition	Example
1	Product ID (Barcode)	Unique identifier of the selected product(s)	Yes	Must exist in the system. Up to ten products may be deleted simultaneously.	893523451234, 89352345123

8. Output data

No	Data fields	Description	Display format	Example
1	Operation result	Notification of successful product deletion or status change.	Text message	Products deleted successfully
2	Error / Validation message	Notification if the operation is invalid (e.g., limit exceeded).	Text message	Cannot delete more than twenty products per day
3	Product status	Status changed to "deactivated" if stock > 0.	Text	Deactivated

9. Postconditions

- The selected products are deleted from the system, or their status is changed to "deactivated" if their stock was greater than zero.
- The software stores a history of products deletion.

4 Supplementary specification

4.1 Functionality

This subsection outlines general business rules and functional constraints shared across the system:

- **Taxation Policy:** *The system shall automatically calculate and apply a 10% Value-Added Tax (VAT) to the original and current prices of all physical media products. Shipping fees are exempt from VAT.*
- **Shipping Fee Rules:** *The system shall dynamically calculate shipping fees based on the total weight and delivery location:*
 - *For Hanoi and Ho Chi Minh City: The base rate is 22,000 VND for the first 3kg.*
 - *For other provinces: The base rate is 30,000 VND for the first 0.5kg.*
 - *Weight Surcharge: An additional 2,500 VND is applied for every subsequent 0.5kg.*
- **Free Shipping Policy:** *Orders with a total product value strictly exceeding 100,000 VND shall receive a shipping discount up to a maximum of 25,000 VND.*
- **Default Payment:** *The system shall default to the VietQR payment method, while providing PayPal Sandbox (Credit Card) as an alternative option.*
- **Automated Notifications:** *The system shall automatically integrate with an Email System to dispatch invoices and transaction receipts upon successful payment or order status updates.*

4.2 Usability

- **Anonymous Checkout:** *The system must allow Customers to browse products, manage their carts, and place orders seamlessly without requiring them to register for an account or log in.*
- **Dynamic Updates:** *The checkout interface must dynamically recalculate and display the shipping fee immediately whenever the customer updates the delivery address.*
- **Error Handling & Recovery:** *The system shall provide clear, actionable error messages (e.g., "Insufficient Stock," "Invalid Credit Card," "Gateway Timeout")*

and must allow users to retry actions or return to previous steps without losing their current session data.

4.3 Reliability

- **Continuous Operation:** *The system must be highly reliable and capable of operating continuously for at least 300 hours without any critical failures or required downtime and can run at least 1 hour after incident.*
- **Availability:** *As an e-commerce platform, the system shall maintain 24/7 availability for customer-facing modules.*
- **Transaction Integrity:** *In the event of a payment gateway timeout or network failure, the system must not drop the user's session. It shall safely abort the transaction, notify the customer, and retain the shopping cart data for a retry.*

4.4 Performance

- **Concurrent Users:** *The system architecture must be robust enough to support up to 1,000 customers accessing and browsing the store simultaneously without significant performance degradation.*
- **Response Time:** *The system shall ensure a maximum response time of 2 seconds in normal hours and 5 seconds in peak hours.*

4.5 Supportability

- **Modular Payment Integration:** *The payment processing module must be designed with loose coupling to allow easy integration of additional payment gateways (e.g., Momo, ZaloPay, domestic bank cards) in future iterations.*
- **Extensible Catalog:** *The database schema for physical media (Books, CDs, DVDs, Newspapers) should utilize inheritance (e.g., a base Media class) to easily support new product types or digital downloads later.*

4.6 Other requirements

- **Security & Access Control:** *The administrative portal must enforce Role-Based Access Control (RBAC). Only authenticated Users can access internal features, and specific actions (like user management) are strictly restricted to Administrators.*
- **External API Dependency:** *The system relies on the availability of the external PayPal Sandbox API and VietQR API.*